

Zhi Wang

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Affiliation **Associate Professor**, Shenzhen International Graduate School, Tsinghua University.

Associate Dean, Institute of Data and Information.

Director, TSIGS – Shine Works InfleXion Lab (Joint Innovation Lab).

Education Background **Ph.D.**, Computer Science and Technology, Tsinghua University, 2008.9–2014.1

B.E., Computer Science and Technology, Tsinghua University, 2004.9–2008.7

Current Research Topics

- Autonomous multimedia intelligence
- Distributed machine learning
- Industrial foundation models
- Embodied industrial intelligence

Achievement Summary

Zhi Wang (S'10-M'14-SM'22) is currently an associate professor at Shenzhen International Graduate School, Tsinghua University. He received his Ph.D. in 2014 and his B.E. in 2008, both from Tsinghua University. His research areas include multimedia networks, mobile cloud computing, and large-scale machine learning systems. He was a recipient of the Natural Science Award of the Ministry of Education (First Prize) in 2017, the National Natural Science Award (Second Prize) in 2018, the Shenzhen Youth Science and Technology Award in 2019, the Technology Invention Award of the Chinese Institute of Electronics (First Prize) in 2020, and the Natural Science Award of Beijing Science and Technology Award (Second Prize) in 2024. In addition, his research won the Best Paper Award of ACM Multimedia, the Best Paper Award of IEEE Transactions on Multimedia, the Outstanding Doctoral Thesis Award of China Computer Federation, the Best Student Paper Award of MMM, and the Best Paper Award of ACM Multimedia, HUMA Workshop. He is an Associate Editor of IEEE TMM and Guest Editor of ACM TIST and JCST. His research has been covered by prestigious technology media, including MIT Technology Review and Synced Review.

Scholarly Research Achievements at a Glance	National Talent Project	1
	National Scientific Research Awards	1
	Provincial and Ministerial Scientific Research Awards	3
	Best Paper Awards	5
	Best Paper Awards Finalists	3
	Grand Challenge Award	1

	Publications Ranked Class A by China Computer Federation (CCF-A)	74
	Refereed Publications in Journals	48
	Refereed Publications in Conference Proceedings	142
	Invited Publications	3
	Refereed Publications in Magazines	5
	Refereed Publications in Books and Chapters	3
	Google Scholar Citations	6754 (as of June 2026)
	H-Index computed by Google Scholar Citations	41 (as of June 2026)
Selected Honors	Natural Science Award of Beijing Science and Technology Award (Second Prize)	2024
	Chinese Institute of Electronics Award (First Prize)	2020
	Shenzhen Youth Science and Technology Award	2019
	National Natural Science Award (Second Class)	2018
	Natural Science Award of Ministry of Education (First Prize)	2017
Best Paper Awards/Finalists	Best Paper Award Finalist , ACM Multimedia 2025	2025
	Grand Challenge Award , 1st Place in Track 1, Vast Vocabulary Object Detection from V3Det Challenge, CVPR	2024
	Best Paper Award , IEEE Transactions on Multimedia	2023
	Best Paper Award , ACM Multimedia, HUMA Workshop 2021	2021
	Best Paper Award Finalist , IEEE ICME 2021	2021
	Best Paper Award Finalist , IEEE/ACM IWQoS 2016	2016
	Best Student Paper Award , MMM 2015	2015
	Outstanding Doctoral Dissertation Award , China Computer Federation (Rank 3 among 10 winners, out of 37 nominees in China)	2014
	Best Paper Award , ACM Multimedia 2012 (only 1 paper receives this award among 4 candidates)	2012
Research Experience	Associate Professor , Tsinghua University, • Tsinghua Shenzhen International Graduate School	2018.12—present
	Assistant Professor , Tsinghua University, • Tsinghua Shenzhen International Graduate School	2014.2—2018.12

PhD Student , Tsinghua University, • Advisor: Prof. Shiqiang Yang • Lab of Multimedia and Networking	2008.9—2014.1
Research Assistant , University of Toronto, • Advisor: Prof. Baochun Li • iQua Group	2012.2—2012.8
Research Assistant , The University of Hong Kong, • Advisor: Prof. Chuan Wu • Networking Research Group	2010.1—2010.7

Publications

Conference papers

1. Xiao Fan, Jingyan Jiang, Zhaoru Chen, Fanding Huang, Xiao Chen, Qinting Jiang, Bowen Zhang, Xing Tang and **Zhi Wang**. “MoETTA: Test-Time Adaptation Under Mixed Distribution Shifts with MoE-LayerNorm,” In Proceedings of Annual AAAI Conference on Artificial Intelligence (AAAI), Singapore, Singapore, January 20-27, 2026. (CCF-A)
2. Jiajun Luo, Yicheng Xiao, Jianru Xu, Yangxiu You, Rongwei Lu, Chen Tang, Jingyan Jiang and **Zhi Wang**. “CompactFusion: Accelerating Parallel Diffusion Model Serving with Residual Compression,” In Proceedings of The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), San Diego, United States, December 2-7, 2025. (CCF-A)
3. Qinting Jiang, Chuyang Ye, Dongyan Wei, Bingli Wang, Yuan Xue, Jingyan Jiang and **Zhi Wang**. “Feature-Based Instance Neighbor Discovery: Advanced Stable Test-Time Adaptation in Dynamic World,” In Proceedings of The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), San Diego, United States, December 2-7, 2025. (CCF-A)
4. Weixiang Zhang, Boxi Li, Shuzhao Xie, Chengwei Ren, Yue Xue and **Zhi Wang**. “Understanding Bias Terms in Neural Representations,” In Proceedings of The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), San Diego, United States, December 2-7, 2025. (CCF-A)
5. Haotian Dong, Jingyan Jiang, Rongwei Lu, Jiajun Luo, Jiajun Song, Bowen Li, Ying Shen and **Zhi Wang**. “Beyond A Single AI Cluster: A Survey of Decentralized LLM Training,” In Proceedings of The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), Suzhou, China, November 4-9, 2025
6. Duo Wu, Jinghe Wang, Yuan Meng, Yanning Zhang, Le Sun and **Zhi Wang**. “CATP-LLM: Empowering Large Language Models for Cost-Aware Tool Planning,” In Proceedings of International Conference on Computer Vision (ICCV), Honolulu, USA, October 19-23, 2025. (CCF-A)
7. Jiajun Luo, Lizhuo Luo, Jianru Xu, Jiajun Song, Rongwei Lu, Chen Tang and **Zhi Wang**. “DICE: Staleness-Centric Optimizations for Parallel Diffusion MoE Inference,” In Proceedings of International Conference on Computer Vision (ICCV), Honolulu, USA, October 19-23, 2025. (CCF-A)

8. Xiaojie Li, Ronghui Li, Shukai Fang, Shuzhao Xie, Xiaoyang Guo, Jiaqing Zhou, Junkun Peng and **Zhi Wang**. “Music-Aligned Holistic 3D Dance Generation via Hierarchical Motion Modeling,” In Proceedings of International Conference on Computer Vision (ICCV), Honolulu, USA, October 19-23, 2025. (CCF-A)
9. Shuzhao Xie, Jiahang Liu, Weixiang Zhang, Shijia Ge, Sicheng Pan, Chen Tang, Yunpeng Bai, Cong Zhang, Xiaoyi Fan and **Zhi Wang**. “SizeGS: Size-aware Compression of 3D Gaussian Splatting via Mixed Integer Programming,” In Proceedings of ACM Multimedia, Dublin, Ireland, October 27-31, 2025. (CCF-A, Best Paper Candidate)
10. Jiacheng Jiang, Yuan Meng, Chen Tang, Han Yu, Qun Li, **Zhi Wang** and Wenwu Zhu. “Quantization Meets OOD: Generalizable Quantization-aware Training from a Flatness Perspective,” In Proceedings of ACM Multimedia, Dublin, Ireland, October 27-31, 2025. (CCF-A)
11. Yuan Xue, Qinting Jiang, Yuan Meng, Xingxuan Zhang, Chen Tang, Jingyan Jiang and **Zhi Wang**. “Dynamic Model Fusion for Multi-Source Test-Time Adaptation,” In Proceedings of The European Conference on Artificial Intelligence (ECAI), Bologna, Italy, October 25-30, 2025.
12. Wei Zhang, Tong Meng, Xianhua Zeng, Changqing Yan, Chao Li, Chenguang Li, Feng Qian, Junfeng Yang, Lei Zhang and **Zhi Wang**. “Harnessing WebRTC for Large-Scale Live Streaming,” In Proceedings of ACM SIGCOMM, Coimbra, Portugal, September 8-11, 2025. (CCF-A)
13. Linghui Zhu, Yiming Li, Haiqin Weng, Yan Liu, Shutao Xia and **Zhi Wang**. “Anti-FT: Towards Practical Deep Leakage from Gradients,” In Proceedings of European Conference on Computer Vision (ICIP), Anchorage, Alaska, September 14-17, 2025
14. Weixiang Zhang, Wei Yao, Shuzhao Xie, Shijia Ge, Chen Tang and **Zhi Wang**. “Expansive Supervision for Neural Radiance Fields,” In Proceedings of In Proceedings of IEEE International Conference on Multimedia and Expo (ICME), Nantes, France, June 30-July 4, 2025
15. Weixiang Zhang, Shuzhao Xie, Chengwei Ren, Chengwei Ren, Siyi Xie, Chen Tang, Shijia Ge, Mingzi Wang and **Zhi Wang**. “EVOS: Efficient Implicit Neural Training via EVolutionary Selector,” In Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, Nashville, TN, USA, June 11-15, 2025 (CCF-A)
16. Lei Chen, Yuan Meng, Chen Tang, Xinzhu Ma, Jingyan Jiang, Xin Wang, **Zhi Wang** and Wenwu Zhu. “Q-DiT: Accurate Post-Training Quantization for Diffusion Transformers,” In Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, Nashville, TN, USA, June 11-15, 2025 (CCF-A)
17. Fanding Huang, Jingyan Jiang, Qingting Jiang, Hebei Li, Faisal Nadeem Khan and **Zhi Wang**. “COSMIC: Clique-Oriented Semantic Multi-space Integration for Robust CLIP Test-Time Adaptation,” In Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, Nashville, TN, USA, June 11-15, 2025 (CCF-A)
18. Rongwei Lu, Yutong Jiang, Jinrui Zhang, Chunyang Zhang, Yifei Zhu, Bin Chen and **Zhi Wang**. “*gamma*-FedHT: Stepsize-Aware Hard-Threshold Gradient Compression in Federated Learning,” In Proceedings of IEEE International Conference on Computer Communications, London, UK 19-22 May 2025 (CCF-A)

19. Shijia Ge, Weixiang Zhang, Shuzhao Xie, Baixu Yan and **Zhi Wang**. “Lung-mix: A Mixup-Based Strategy for Generalization in Respiratory Sound Classification,” In Proceedings of 2025 IEEE International Conference on Acoustics, Speech and Signal Processing Hyderabad, India, April 06-11 2025
20. Baixu Yan, Shijia Ge, Meizi Lu, Weixiang Zhang, Shuzhao Xie and **Zhi Wang**. “PulmoScan: A Practical Pulmonary Disease Pre-Screening System,” In Proceedings of IEEE International Conference on Acoustics, Speech, and Signal Processing, Hyderabad, India April 06-11 2025.
21. Weixiang Zhang, Shuzhao Xie, Chengwei Ren, Shijia Ge, Mingzi Wang and **Zhi Wang**. “Enhancing Implicit Neural Representations via Symmetric Power Transformation,” In Proceedings of AAAI Conference on Artificial Intelligence, Philadelphia, USA, February 25-March 4, 2025 (CCF-A)
22. Mingzi Wang, Yuan Meng, Chen Yan, Weixiang Zhang, Yijian Qin, Yang Yao, Yingxin Li, Tongtong Feng, Xin Wang, Xun Guan, **Zhi Wang** and Wenwu Zhu. “JAQ: Joint Efficient Architecture Design and Low-Bit Quantization with Hardware-Software Co-Exploration,” In Proceedings of AAAI Conference on Artificial Intelligence, Philadelphia, USA, February 25-March 4, 2025 (CCF-A)
23. Shijia Ge, Weixiang Zhang, Shuzhao Xie, Baixu Yan and **Zhi Wang**. “Lung-mix: A Mixup-Based Strategy for Generalization in Respiratory Sound Classification,” In Proceedings of 2025 IEEE International Conference on Acoustics, Speech and Signal Processing Hyderabad, India, April 06-11 2025
24. Baixu Yan, Shijia Ge, Meizi Lu, Weixiang Zhang, Shuzhao Xie and Zhi Wang. “PulmoScan: A Practical Pulmonary Disease Pre-Screening System,” In Proceedings of IEEE International Conference on Acoustics, Speech, and Signal Processing, Hyderabad, India April 06-11 2025.
25. Rongwei Lu, Yutong Jiang, Jinrui Zhang, Chunyang Zhang, Yifei Zhu, Bin Chen and **Zhi Wang**. “ γ -FedHT: Stepsize-Aware Hard-Threshold Gradient Compression in Federated Learning,” In Proceedings of IEEE International Conference on Computer Communications, London, UK 19-22 May 2025 (CCF-A)
26. Zijie Fang, Yifeng Wang, **Zhi Wang**, Jian Zhang, Xiangyang Ji and Yongbing Zhang. “MamMIL: Multiple Instance Learning for Whole Slide Images with State Space Models,” In Proceedings of IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Lisbon, Portugal, December 3-6, 2024
27. Shuzhao Xie, Weixiang Zhang, Chen Tang, Yunpeng Bai, Rongwei Lu, Shijia Ge and **Zhi Wang**. “MesonGS: Post-training Compression of 3D Gaussians via Efficient Attribute Transformation,” In Proceedings of European Conference on Computer Vision (ECCV), Milano, Italy, September 29-October 4, 2024 (CCF-A)
28. Jiajun Song, Jiajun Luo, Rongwei Lu, Shuzhao Xie, Bin Chen and **Zhi Wang**. “A Joint Approach to Local Updating and Gradient Compression for Efficient Asynchronous Federated Learning,” In Proceedings of International European Conference on Parallel and Distributed Computing (Euro-Par), Madrid, Spain, August 26-30, 2024.
29. Duo Wu, Xianda Wang, Yaqi Qiao, **Zhi Wang**, Junchen Jiang, Shuguang Cui and Fangxin Wang. “NetLLM: Adapting Large Language Models for Networking,” In Proceedings of (ACM SIGCOMM), Sydney, Australia, August 4-8, 2024 (CCF-A)

30. Jinmin Li, Tao Dai, Yaohua Zha, Yilu Luo, Longfei Lu, Bin Chen, **Zhi Wang**, Shutao Xia and Jingyun Zhang. “Invertible Residual Rescaling Models,” In Proceedings of International Joint Conference on Artificial Intelligence (IJ-CAI), Jeju, August 3-9, 2024 (**CCF-A**)
31. Mingzhou Wu, Shiqi Dai, Han Hu and **Zhi Wang**. “Collaborative Edge Caching in LEO Satellites Networks: A MAPPO Based Approach,” In Proceedings of IEEE International Conference on Multimedia and Expo (ICME), Niagara Falls, Canada, July 15-19, 2024
32. Chen Tang, Yuan Meng, Jiacheng Jiang, Shuzhao Xie, Rongwei Lu, Xinzhu Ma, **Zhi Wang** and Wenwu Zhu. “Retraining-free Model Quantization via One-Shot Weight-Coupling Learning,” In Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, June 17-21, 2024 (**CCF-A**)
33. Yuling Zhang, Anpeng Wu, Kun Kuang, Liang Du, Zixun Sun and **Zhi Wang**. “Stable Heterogeneous Treatment Effect Estimation across Out-of-Distribution Populations,” In Proceedings of IEEE International Conference on Data Engineering (ICDE), Utrecht, Netherlands, May 13-16, 2024 (**CCF-A**)
34. Qun Li, Yuan Meng, Chen Tang, Jiacheng Jiang and **Zhi Wang**. “Investigating the Impact of Quantization on Adversarial Robustness,” In Proceedings of Workshop on Practical Machine Learning for Developing Countries (PML4DC), ICLR, Vienna Austria, May 3-11, 2024
35. Shiqi Dai, Xuanyu Zhu, Naiqi Li, Tao Dai and **Zhi Wang**. “Procedural Level Generation with Diffusion Models from a Single Example,” In Proceedings of Annual AAAI Conference on Artificial Intelligence, Vancouver, NO, Canada, February 20–27, 2024 (**CCF-A**)
36. Yaohua Zha, Huizhen Ji, Jinmin Li, Rongsheng Li, Tao Dai, Bin Chen, **Zhi Wang** and Shutao Xia. “Towards Compact 3D Representations via Point Feature Enhancement Masked Autoencoders,” In Proceedings of Annual AAAI Conference on Artificial Intelligence, Vancouver, NO, Canada, February 20–27, 2024 (**CCF-A**)
37. Taolin Zhang, Sunan He, Tao Dai, Bin Chen, **Zhi Wang** and Shutao Xia. “Vision-Language Pre-training with Object Contrastive Learning for 3D Scene Understanding,” In Proceedings of Annual AAAI Conference on Artificial Intelligence, Vancouver, NO, Canada, February 20–27, 2024 (**CCF-A**)
38. Bin Huang, Jiaqian Yu, Yiwei Chen, Siyang Pan, Qiang Wang and **Zhi Wang**. “BadTrack: A Poison-Only Backdoor Attack on Visual Object Tracking,” In Proceedings of NeurIPS, 2023, New Orleans, NO, USA, December 10–16, 2023 (**CCF-A**)
39. Zheming Yang, Wen Ji, Qi Guo and **Zhi Wang**. “JAVP: Joint-Aware Video Processing with Edge-Cloud Collaboration for DNN Inference,” In Proceedings of ACM Multimedia, Ottawa, Canada, October 29–November 3, 2023 (**CCF-A**)
40. Chen Tang, Kai Ouyang, Yunpeng Bai, Yuan Meng, **Zhi Wang** and Wenwu Zhu. “SEAM: Searching Transferable Mixed-Precision Quantization Policy through Large Margin Regularization,” In Proceedings of ACM Multimedia, Ottawa, Canada, October 29–November 3, 2023 (**CCF-A**)
41. Hang Guo, Tao Dai, Mingyan Zhu, Guanghao Meng, Bin Chen, **Zhi Wang** and Shutao Xia. “One-stage Low-resolution Text Recognition with High-resolution Knowledge Transfer,” In Proceedings of ACM Multimedia, Ottawa, Canada, October 29–November 3, 2023 (**CCF-A**)

42. Bin Huang and **Zhi Wang**. “Efficient Any-Target Backdoor Attack with Pseudo Poisoned Samples,” In Proceedings of IEEE ICIP, Kuala Lumpur, Malaysia, October 8–11, 2023
43. Mingqing Wang, Jiawei Li, Bin Chen, Shutao Xia and **Zhi Wang**. “Unsupervised Anomaly Detection with Local-Sensitive VQVAE and Global-Sensitive Transformers,” In Proceedings of IEEE ICIP, Kuala Lumpur, Malaysia, October 8–11, 2023
44. Jianghui Zhang, Yuanyuan Wang, Lina Guo, Jixiang Luo, Tongda Xu, Yan Wang, **Zhi Wang** and Hongwei Qin. “Unified learning-based lossy and lossless JPEG recompression,” In Proceedings of IEEE ICIP, Kuala Lumpur, Malaysia, October 8–11, 2023
45. Yaohua Zha, Jinpeng Wang, Tao Dai, Bin Chen, **Zhi Wang** and Shutao Xia. “Instance-aware Dynamic Prompt Tuning for Pre-trained Point Cloud Models,” In Proceedings of International Conference on Computer Vision, Paris, France, 2-6 October 2023 (**CCF-A**)
46. Chen Tang, Li Lina Zhang, Huiqiang Jiang, Jiahang Xu, Ting Cao, Quanlu Zhang, Yuqing Yang, **Zhi Wang** and Mao Yang. “ElasticViT: Conflict-aware Supernet Training for Deploying Fast Vision Transformer on Diverse Mobile Devices,” In Proceedings of International Conference on Computer Vision, Paris, France, 2-6 October 2023 (**CCF-A**)
47. Hao Fang, Bin Chen, Xuan Wang, **Zhi Wang** and Shutao Xia. “GIFD: A Generative Gradient Inversion Method with Feature Domain Optimization,” In Proceedings of International Conference on Computer Vision, Paris, France, 2-6 October 2023 (**CCF-A**)
48. Yinan Mao, Bowei He, Shiji Zhou, Chen Ma and **Zhi Wang**. “Collaborative Edge Caching: a Meta Reinforcement Learning Approach with Edge Sampling,” In Proceedings of IEEE ICME, Brisbane, Australia, 10-15 July 2023
49. Dieli Hu, Wen Ji and **Zhi Wang**. “Multi-stream Adaptive Offloading of Joint Compressed Video Streams, Feature Streams, and Semantic Streams in Edge Computing Systems,” In Proceedings of IEEE ICME, Brisbane, Australia, 10-14 July 2023
50. Yinan Mao, Shiji Zhou, Haochen Liu, **Zhi Wang** and Wenwu Zhu. “Dynamic Edge Caching via Online Meta-RL,” In Proceedings of IEEE IJCNN, Queensland, Australia, 18-23 June 2023
51. Xiaochan Wang and **Zhi Wang**. “Personalized Federated Learning via Knowledge Sharing-based Model Structure Adaption,” In Proceedings of IEEE IJCNN, Queensland, Australia, 18-23 June 2023
52. Rongwei Lu, Jiajun Song, Bin Chen, Laizhong Cui and **Zhi Wang**. “DAGC: Data-aware Adaptive Gradient Compression,” In Proceedings of IEEE INFOCOM, New York, USA, 17-20 May 2023 (**CCF-A**)
53. Zhenyang Li, Yancheng Dong, Chen Gao, Yizhou Zhao, Dong Li, Jianye Hao, Kai Zhang, Yong Li and **Zhi Wang**. “Breaking Filter Bubble: A Reinforcement Learning Framework of Controllable Recommender System,” In Proceedings of The ACM Web Conference 2023, Austin, Texas, USA, 30 April-4 May 2023 (**CCF-A**)
54. Jinmin Li, Tao Dai, Mingyan Zhu, Bin Chen, **Zhi Wang** and Shutao Xia. “FSR: A General Frequency-oriented Framework to Accelerate Image Super-resolution Networks,” In Proceedings of Thirty-Seventh AAAI Conference on Artificial Intelligence, Washington, DC, USA, 7-14 February 2023 (**CCF-A**)

55. Zijie Fang, Yang Chen, Yifeng Wang, **Zhi Wang**, Xiangyang Ji and Yongbing Zhang. “Weakly-Supervised Semantic Segmentation for Histopathology Images Based on Dataset Synthesis and Feature Consistency Constraint,” In Proceedings of Thirty-Seventh **AAAI** Conference on Artificial Intelligence, Washington, DC, USA, 7-14 February 2023 (**CCF-A**)
56. Xiaohan Lan, Yitian Yuan, Hong Chen, Xin Wang, Zequn Jie, Lin Ma, **Zhi Wang** and Wenwu Zhu. “Curriculum Multi-Negative Augmentation for De-biased Video Grounding,” In Proceedings of Thirty-Seventh **AAAI** Conference on Artificial Intelligence, Washington, DC, USA, 7-14 February 2023 (**CCF-A**)
57. Yiyun Lu, Yifei Zhu and **Zhi Wang**. “Personalized 360-Degree Video Streaming: A Meta-Learning Approach,” In Proceedings of **ACM Multimedia**, Lisbon, Portugal, 10-14 October 2022 (**CCF-A**)
58. Chen Tang, Haoyu Zhai, Kai Ouyang, **Zhi Wang**, Yifei Zhu and Wenwu Zhu. “Arbitrary Bit-width Network: A Joint Layer-Wise Quantization and Adaptive Inference Approach,” In Proceedings of **ACM Multimedia**, Lisbon, Portugal, 10-14 October 2022 (**CCF-A**)
59. Chen Tang, Kai Ouyang, **Zhi Wang**, Yifei Zhu, Wen Ji, Yaowei Wang and Wenwu Zhu. “Mixed-Precision Neural Network Quantization via Learned Layer-wise Importance,” In Proceedings of European Conference on Computer Vision (ECCV), Tel-Aviv, Israel, 23-27 October 2022
60. Xinjie Yuan, Mingzhou Wu, **Zhi Wang**, Yifei Zhu, Ming Ma, Junjian Guo, Zhili Zhang and Wenwu Zhu. “Understanding 5G Performance for Real-world Services: a Content Provider’s Perspective,” In Proceedings of (**ACM SIGCOMM**), Amsterdam, the Netherlands, 22-26 August 2022 (**CCF-A**)
61. Ruoyu Zhang, Yutao Zhou, Fangxin Wang and **Zhi Wang**. “Maxim: DRL-Based Cross-camera Streaming Configuration for Real-time Video Analytics,” In Proceedings of IEEE International Conference on Multimedia and Expo (ICME), Taipei, Taiwan, 18-22 July 2022.
62. Zhaoliang He, Yuan Meng, Chen Yan, **Zhi Wang**, Wenwu Zhu, Chenyang Guo and Zhibo Chen. “AdaConfigure: Reinforcement Learning-based Adaptive Configuration for Video Analytics Services,” In Proceedings of International Conference on MultiMedia Modeling (MMM), Phu Quoc, Vietnam, 6-10 June 2022.
63. Cheng Chen, Yichun Yin, Lifeng Shang, Xin Jiang, Yujia Qin, Fengyu Wang, **Zhi Wang**, Xiao Chen, Zhiyuan Liu and Qun Liu. “bert2BERT: Towards Reusable Pretrained Language Models,” In Proceedings of Annual Meeting of the Association for Computational Linguistics (**ACL**), Dublin, 22-27 May 2022. (**CCF-A**)
64. Shuzhao Xie, Yuan Xue, Yifei Zhu and **Zhi Wang**. “Cost Effective MLaaS Federation: A Combinatorial Reinforcement Learning Approach,” In Proceedings of **IEEE INFOCOM**, Virtual Conference, 2-5 May 2022. (**CCF-A**)
65. Lei Zhang, Yuqing Zhang, Ximing Wu, Fangxin Wang, Laizhong Cui, **Zhi Wang** and Jiangchuan Liu. “Batch Adaptive Streaming for Video Analytics,” In Proceedings of **IEEE INFOCOM**, Virtual Conference, 2-5 May 2022. (**CCF-A**)
66. Shiji Zhou, Han Zhao, Shanghang Zhang, Lianzhe Wang, Heng Chang, **Zhi Wang** and Wenwu Zhu. “Online Continual Adaptation with Active Self-Training,” In Proceedings of International Conference on Artificial Intelligence and Statistics, Virtual Conference, 28-30 March 2022.

67. Jingyan Jiang, Ziyue Luo, Chenghao Hu, Zhaoliang He, **Zhi Wang**, Shutao Xia and Chuan Wu. “Joint Model and Data Adaptation for Cloud Inference Serving,” In Proceedings of IEEE Real-Time Systems Symposium (**RTSS**), Dortmund, Germany, 7–10 December 2021. (**CCF-A**)
68. Yitian Yuan, Xiaohan Lan, Xin Wang, Long Chen, **Zhi Wang** and Wenwu Zhu. “A Closer Look at Temporal Sentence Grounding in Videos: Dataset and Metric,” In Proceedings of **ACM Multimedia**, HUMA Workshop, Chengdu, China, 20–24 October 2021. (**Best Paper Award**)
69. Tao Dai, Yalei Lv, Bin Chen, **Zhi Wang**, Zexuan Zhu and Shutao Xia. “Mix-order Attention Networks for Image Restoration,” In Proceedings of **ACM Multimedia**, Chengdu, China, 20-24 October 2021. (**CCF-A**)
70. Chenglei Wu, **Zhi Wang** and Lifeng Sun. “A Spherical Mixture Model Approach for 360 Video Virtual Cinematography,” In Proceedings of IEEE International Conference on Image Processing (ICIP), Anchorage, Alaska, USA, 19-22 September 2021.
71. Cheng Chen, Yichun Yin, Lifeng Shang, **Zhi Wang**, Xin Jiang, Xiao Chen and Qun Liu. “Extract then Distill: Efficient and Effective Task-Agnostic BERT Distillation,” In Proceedings of The International Conference on Artificial Neural Networks, Fully Online, 14-17 September 2021.
72. Wenjie Zhang and **Zhi Wang**. “Blurred Image Recognition: A Joint Motion Deblurring and Classification Loss-Aware Approach,” In Proceedings of The International Conference on Artificial Neural Networks, Fully Online, 14-17 September 2021.
73. Jian Zhao, Xin Wu, Yan Zhang, Yu Wu and **Zhi Wang**. “A Blockchain Based Decentralized Gradient Aggregation Design for Federated Learning,” In Proceedings of The International Conference on Artificial Neural Networks, Fully Online, 14-17 September 2021.
74. Chenglei Wu, **Zhi Wang** and Lifeng Sun. “PAAS: A Preference-Aware Deep Reinforcement Learning Approach for 360 Video Streaming,” In Proceedings of ACM NOSSDAV, Istanbul, Turkey, September 2021.
75. Miao Zhang, Fangxin Wang, Yifei Zhu, Jiangchuan Liu and **Zhi Wang**. “Towards Cloud-Edge Collaborative Online Video Analytics with Fine-Grained Serverless Pipelines,” In Proceedings of ACM MMSys, Istanbul, Turkey, September 2021.
76. Kun Yi, **Zhi Wang** and Lifeng Sun. “A Dataset for Exploring Gaze Behaviors in Text Summarization,” In Proceedings of ACM NOSSDAV, Istanbul, Turkey, September 2021.
77. Haopeng Yan, Zeming Chen, **Zhi Wang** and Wenwu Zhu. “DRL-based Collaborative Edge Content Replication with Popularity Distillation,” In Proceedings of IEEE ICME 2021, Shenzhen, China, July 5-9, 2021.
78. Zhiwei Hao, Yong Luo, **Zhi Wang**, Han Hu and Jianping An. “Model Compression via Collaborative Data-free Knowledge Distillation for Edge Intelligence,” In Proceedings of IEEE ICME 2021, Shenzhen, China, July 5-9, 2021. (**Best Paper Award Finalist**)
79. Jiahui Ye, Zichun Li, **Zhi Wang**, Zhuobin Zheng, Han Hu and Wenwu Zhu. “Joint Cache Size Scaling and Replacement Adaptation for Small Content Providers,” In Proceedings of IEEE **INFOCOM** 2021, Virtual Conference, China, May 10-13, 2021. (**CCF-A**)

80. Kun Yi, Yu Guo, **Zhi Wang**, Lifeng Sun and Wenwu Zhu. "Personalized Text Summarization Based on Gaze Patterns," In Proceedings of IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR), Shenzhen, Guangdong, China August 6-8, 2020.
81. Peng Zhang, Yuwei Zhang, Shan Huang and **Zhi Wang**. "Fine-grained Garment Parsing: a Body Generation Approach," In Proceedings of IEEE International Conference on Multimedia and Expo (ICME) 2020, London, UK, July 6-10, 2020.
82. Yuwei Zhang, Peng Zhang, Chun Yuan, and **Zhi Wang**. "Texture and Shape biased Two-Stream Networks for Clothing Classification and Attribute Recognition," In Proceedings of **IEEE CVPR**, Virtual, June 14-19, 2020. (CCF-A)
83. Cong Zhang, Jiangchuan Liu, **Zhi Wang** and Lifeng Sun. "Look Ahead at the First-mile in Livecast with Crowdsourced Highlight Prediction," In Proceedings of **IEEE INFOCOM**, Beijing, China, April 27-30, 2020. (CCF-A)
84. Chenglei Wu, Ruixiao Zhang, **Zhi Wang** and Lifeng Sun. "A Spherical Convolution Approach for Learning Long Term Viewport Prediction in 360 Immersive Video," In Proceedings of **AAAI**, New York, NY, USA, February 7-12, 2020. (CCF-A)
85. Hongshan Li, Yu Guo, **Zhi Wang**, Shutao Xia and Wenwu Zhu. "AdaCompress: Adaptive Compression for Online Computer Vision Services," In Proceedings of **ACM Multimedia**, Nice, France. October 21-25, 2019. (CCF-A)
86. Shan Huang, **Zhi Wang**, Laizhong Cui, Yong Jiang and Rui Gao. "Fine-grained Fitting Experience Prediction: A 3D-slicing Attention Approach," In Proceedings of **ACM Multimedia**, Nice, France, October 21-25, 2019. (CCF-A)
87. Chenghao Hu, Jingyan Jiang, **Zhi Wang**. "Decentralized Federated Learning: A Segmented Gossip Approach" In Proceedings of **IJCAI Workshop on Federated Machine Learning**, Macau, China, August 10-16, 2019. (CCF-A)
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1. **Zhi Wang**, Jiangchuan Liu and Wenwu Zhu. “Edge Intelligence Empowered Immersive Media,” In IEEE MultiMedia, vol. 30, no. 2, pp. 8–17, April-June 2023.
2. Lifeng Sun, Ming Ma, Wen Hu, Haitian Pang and **Zhi Wang**. “Beyond 1 Million Nodes—Crowdsourced Video CDN: Architecture, Technology, and Economy,” IEEE MultiMedia, June 2017.
3. Lei Zhang, Liting Zhao, **Zhi Wang**, Jiangchuan Liu, “WiFi Networks in Metropolises: From Access Point and User Perspectives,” IEEE Communications Magazine, vol. 55, no. 5, pp. 42–48, May 2017.
4. **Zhi Wang**, Jiangchuan Liu, Wenwu Zhu, “Social Video Delivery: Challenges, Approaches and Directions,” IEEE Network Magazine, vol. 30, no. 5, pp. 35–39, September 2016.
5. Wenwu Zhu, Peng Cui, **Zhi Wang**, Gang Hua. “Multimedia Big Data Computing,” IEEE MultiMedia, vol. 22, no. 3, pp. 96–c3, July 2015.

Invited Talks

“A Preliminary Study on Industrial Intelligence for Edge Computing Network Environment”

- GAIR Summit, Keynote, Shenzhen, China, December 12-13, 2025
- Tsinghua University Anniversary Event, Keynote, Shenzhen, China, May 10, 2025

“Large Model Training and Inference Acceleration and Tool Collaboration for Resource-Constrained Scenarios”

- Tencent Global Digital Ecosystem Summit, Keynote, Shenzhen, China, September 16, 2025

“Computation-Aware Distributed Machine Learning and LLM Deployment”

- Tsinghua Alumni Sanchuang Entrepreneurship Competition, Keynote, Zhenzhou, China, August 15, 2024

“Edge Intelligence Empowered Immersive Media”

- ByteDance, Shenzhen, China, June 5, 2024
- SkyWorth, Shenzhen, China, May 30, 2024
- Shenzhen University, Shenzhen, China, May 5, 2024
- The 17th China Electronics and Information Conference, Ningbo, China, April 28, 2024
- China High-Tech Fair at Shenzhen, Shenzhen, China, November 16, 2023

- Southern University of Science and Technology, Shenzhen, China, November 10, 2023
- Intel China Academia Forum 2023, Nanjing, China, August 16-17, 2023
- SIGS, Tsinghua University, February 25, 2023

“Understanding 5G Performance for Real-world Services: a Content Provider’s Perspective”

- HUAWEI, December 23, 2022

“Metaverse: An Multimedia Edge Computing Perspective”

- Intel, Online, December 2, 2022

“Data-driven Multimedia Edge Networks”

- ICASSP, Shenzhen, October 28, 2022
- Shenzhen International Graduate School, Shenzhen, September 15, 2022

“From Cross-domain Multimedia Content Delivery to Multimedia Edge Computing”

- Shenzhen University, Shenzhen, December 3, 2021

“Multimedia Edge Network and Content Delivery”

- SIGS, Tsinghua University, Shenzhen, July 15, 2021
- Chinese Institute of Electronics Multimedia Network Society Forum, Chongqing, March 26, 2018

“Behavior-driven Social Content Delivery”

- IEEE CAS Forum, Zhengzhou, July 21, 2016

“Building Better Edge Networks: A Data-driven Approach”

- Shenzhen University, Shenzhen, December 8, 2015
- The University of Hong Kong, Hong Kong, October 2, 2015
- Young Research Forum, ACM ICIMCS 2015, Zhangjiajie, China, August 21, 2015
- Big-data Innovation Forum, Shenzhen International Graduate School, Shenzhen, China, May 24, 2015

“Strategies for Social Content Delivery”

- Tsinghua-Huawei Network Workshop, Shenzhen International Graduate School, Shenzhen, China, December 18, 2014
- Tencent Share Day, Tencent, Shenzhen, China, April 12, 2014

“From Data-Driven Social Media Deployment to Big-Data Cloud Service”

- Job Talk, at Shenzhen International Graduate School, Tsinghua University, Shenzhen, China, March 27, 2013

“Data-Driven Approach to Studying Social Media Deployment”

- Tsinghua-CUHK Joint Lab, at Shenzhen International Graduate School, Tsinghua University, Shenzhen, China, December 20, 2012

“Cloud-Based Social Application Deployment Using Local Processing and Global Distribution”

- ACM CoNEXT'2012, at Nice, France, December 13, 2012

“TCP Optimization for Content Provider”

- Tencent CDN Group, at Tencent Shenzhen, China, December 6, 2012

“Social-Aware Delivery of Online Multimedia Content”

- Outstanding PhD forum, at HHME 2012, Guangzhou, China, November 24, 2012

“Propagation-Based Social-Aware Replication for Social Video Contents”

- Huawei, at Shenzhen, China, November 22, 2012
- Best paper presentation, at ACM Multimedia 2012, Nara, Japan, October 30, 2012

Professional Service Activities

Journal Editorships

- Associate Editor, IEEE Transactions on Multimedia, 2019–2023
- Associate Editor, IEEE Open Journal of Communications Society, 2019–2023
- Guest Editor, Future Generation Computer Systems, Special issue on Server-less Computing in the Cloud-to-Edge Continuum, 2023
- Guest Editor, ACM Transactions on Intelligent Systems and Technology (ACM TIST), Special issue on Data-driven Intelligence for Wireless Networking, 2016–2017
- Guest Editor, Special Section on Fog Computing, Journal of Computer and Technology (JCST), 2016–2017
- Guest Editor, Special Section on Networking and Distributed Computing for Big Data, Journal of Computer and Technology (JCST), 2015–2016
- Leading Guest Editor, Special Issue on Computer Network Design for Social Media, China Communications, 2015–2016

Conference/IG Chairs

- Industrial Chair, IVMS2026, Shenzhen, China, June 2026
- Area Chair, ACM Multimedia, Dublin, Ireland, October 2025
- Chair, Communication Software and Multimedia Symposium, IEEE ICC, Montreal Canada, June 2025
- Chair, IEEE ComSoc Multimedia Big Data Special Interest Group, 2017–2018
- Co-Chair, IEEE ICME MuSIC Workshop 2017
- Track Chair, IEEE PhoneCom 2014

Membership in Technical Program Committees

- ACM Multimedia (MM) 2014–
- CIKM 2019–
- ICDM 2018–
- ICLR 2023–
- CVPR 2023–
- ACM Multimedia Systems Conference (MMSys) 2014–
- IEEE International Conference on Mobile Ad hoc and Sensor Systems (MASS) 2015–

- IJCAI 2020, 2023
- ICCCRI 2016
- IEEE International Conference on Computer Communications (INFOCOM) 2015
- IEEE Big Data Congress 2015
- IEEE International Conference on Networking, Architecture, and Storage (NAS) 2015, 2014
- Pacific-Rim Conference on Multimedia (PCM) 2014
- IEEE International Conference on Cyber, Physical and Social Computing (CPSCoM) 2014
- IEEE International Conference on Communications (ICC) 2014
- IEEE International Conference on Multimedia and Expo (ICME) 2013

Reviewer

- Journal: IEEE TMM, IEEE TCSVT, IEEE/ACM ToN, ACM TOMCCAP, IEEE TPDS, IEEE TMC, IEEE JSAC, IEEE ACCESS, Springer ComCom
- Conference: IEEE ICDM, IEEE VCIP, ACM Multimedia, ACM MMSys, NAS, CPSCoM, IEEE INFOCOM, ACM MOBICOM, IEEE IWQoS, IEEE ICNP, IEEE ICME, IEEE GLOBECOM, IEEE GLOBECOM

Services at Tsinghua University

- Associate Dean (2024.8–)
- Postgraduate Affairs Working Group (2023.9–2024.8)
- Dean Assistant (2023.1–2024.7)

Other Services

- Voting Member, IEEE Standards Association, Digital Retina Systems Working Group (3161 WG), 2022–

EDICS Of Interest

Ultra Low Delay Next Generation Multimedia Communication and Networking; Multimedia Streaming; Multimedia Networking and Processing in the Cloud and Data Centers; Joint Media and Multimodal Processing; Virtual Reality, Augmented Reality, and Virtual Environments; Deep Learning for Multimedia Analysis; Deep Learning for Multimedia Processing;

Teaching Experience

Teaching Awards

Seminars on Data Science and Applications, selected as one of the Shenzhen “Quality Shared Courses”. 2024

Advanced Big Data Systems, selected as one of the initial online demonstration courses for graduate students for professional engineering degrees. 2023

Advanced Big Data Systems, recognized as a first-class national undergraduate course. 2020

Distributed Machine Learning, ranked in the top 5% by the teaching evaluations at Tsinghua University. 2020

Advanced Big Data Systems, designated a National Quality Online Open Course. 2020

Smart Teaching Star Award, the Ministry of Education's Online Education Center. 2019

Outstanding Employee Award, Tsinghua University. 2017

The Teaching Pioneer Award, the Ministry of Education's Online Education Center. 2016

Courses at Tsinghua University

64100033, **Foundations for Big Data Systems**, Fall 2014–
• 16-week course on distributed systems and networks for big-data storage, processing and visualization.

60250131, **Seminars on Data Science and Applications**, Fall 2015–
• 8-week course on frontier seminars of data science.

85990072, **Distributed Machine Learning**, Spring 2021–
• 16-week course on distributed machine learning systems and distributed optimization algorithms.

82558001, **Literature Searching and Academic Writing**, Spring 2021–
• 8-week course on helping graduate students to gain the capabilities for research by starting with literature search and writing skills.

80240683, **Big Data Visualization**, Spring/Fall 2016–2020
• 16-week course on big data visualization technologies.

86000152, **Foundations for Big-data Analytics**, Spring/Fall 2015–2018
• 16-week course on introduction to big-data analytical techniques.

Online Course (MOOC)

Advanced Big Data Systems, Fall 2016–
• The online version of Foundations for Big Data Systems.

Research Grants

Basic Research Grants

1. Role: PI, Source: NSFC Key Project, Period: 2025-2027, Project: Industrial Large Model Architecture and Algorithms for Intelligent Manufacturing, Amount: 2.3M (CNY)

2. Role: PI, Source: NSFC General Project, Period: 2025-2028, Project: Efficient Distributed Machine Learning in Resource-Constrained Scenarios, Amount: 500K (CNY)
3. Role: PI, Source: National Key Basic Research Program (Project), Period: 2023-2026, Project: Research and Application of a Platform With Post-market Services After Equipment Manufacturing, Amount: 1.8M (CNY)
4. Role: PI, Source: Shenzhen Basic Research Key Project, Period: 2023-2026, Project: Research on Metaverse Content Generation and Analysis, Amount: 2.0M (CNY)
5. Role: PI, Source: National High-level Talent Special Support Plan, Period: 2020-2023, Project: Multimedia Edge Computing, Amount: 1.7M (CNY)
6. Role: PI, Source: NSFC General Project, Period: 2019-2022, Project: Data-driven Edge Network Content Distribution Strategy, Amount: 620K (CNY)
7. Role: PI, Source: Shenzhen Outstanding Talent Project, Period: 2021-2024, Project: Multimedia Network and Edge Content Distribution, Amount: 2M (CNY)
8. Role: PI, Source: Shenzhen Basic Research Project, Period: 2019-2021, Project: Online Virtual Reality Content Transmission Strategy Based on User Behavior Prediction, Amount: 500K (CNY)
9. Role: PI, Source: NSFC Youth Project, Period: 2015-2017, Project: Theory and Methodology for Cloud Deployment of Mobile Social Media Services, Amount: 280K (CNY)
10. Role: PI, Source: Shenzhen Basic Research Project, Period: 2014-2016, Project: Strategies of Mobile Cloud Deployment for Mobile Social Media Services, Amount: 300K (CNY)
11. Role: Co-PI, Source: Shenzhen Basic Research Major Special Project, Period: 2025-2027, Project: Humanoid Standard Behavior Library Data Collection and Embodied Intelligent Robot Application Technology Development, Amount: 5.0M (CNY)
12. Role: Co-PI, Source: NSFC Major Project, Period: 2016-2020, Project: Theory and key Technology of User-experience-guaranteed Mobile Video Service Distribution, Amount: 560K (CNY)
13. Role: Co-PI, Source: National Key Basic Research Program, Period: 2015-2018, Project: Three-dimensional Spatial Big Data Computing Theory and Method for Urban Management, Amount: 870K (CNY)

Applied Research Grants

1. Role: PI, Source: Shine Works, Tsinghua-shine Works Joint Innovation lab, Period: 2018-2026, Project: Joint Innovation lab, Amount: 10M (CNY)
2. Role: PI, Source: China National Depository and Clearing Corporation Limited, Period: 2024-2025, Project: Financial Instruction Fine-tuning Dataset Enhancement Project for CCDC, Amount: 1.6M (CNY)
3. Role: PI, Source: Huawei Technologies Co., Ltd., Period: 2025-2026, Project: Collaborative Project on Embodied Data Synthesis Technology for Spatial Perception Understanding and Motion Trajectory Optimization, Amount: 1.33M (CNY)

4. Role: PI, Source: ByteDance, Period: 2024-2025, Project: Research on Multi-media Networks with LLM, Amount: 450K (CNY)
5. Role: PI, Source: Kuaishou, Period: 2021-2026, Projects: White-boxed QoS-QoE Relationship Mining, LLM-based QoE Optimization, Amount: 1.7M (CNY)
6. Role: PI, Source: Tencent, Period: 2020-2021, Project: Large-scale Distributed Training and AutoML Framework, Amount: 200K (CNY)
7. Role: PI, Source: Epoque, Period: 2017-2019, Project: 3D Generation and Data Mining Research, Amount: 1M (CNY)
8. Role: PI, Source: Jiuzhou, Period: 2016-2018, Project: Research and Development of key Technologies for Content Distribution of Smart TV Terminals Based on Data Driven, Amount: 350K (CNY)
9. Role: PI, Source: Belle, Period: 2016-2017, Project: big Data Mining for Manufacturing, Amount: 1.273M (CNY)
10. Role: PI, Source: Tencent, Period: 2015-2016, Project: WIFI Knowledge Graph System Research, Amount: 300K (CNY)
11. Role: PI, Source: Tencent, Period: 2014-2015, Project: Research on Edge Content Transport Architecture, Strategy and Potential Performance, Amount: 200K (CNY)